TALH: A Tangible User Interface and Hand Gestures-Based Tool for Enhancing Animal Learning in Preschool

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***Abstract*—Child interaction and behavior analysis within learning environments are crucial to improving the learning process and experience. In this paper, we employ a comparitive study between different interaction techniques, such as using tangible user interface objects (TUIOs) and gesture-based interaction to teach preschool children about animals in different habitats, monitoring their learning process, and focusing on maximizing engagement and usability within the learning context. We present TALH, an interactive animal habitat simulator for animals in three different habitats. Features such as facial identification, eye gaze tracking, and emotion analysis were implemented to provide deeper insights into child behavior and preferences during the interactive session. Reports for teachers are generated in real-time at the end of the experiment for further analysis. The experimental study involves 10 child participants performing specific tasks using both interaction methods in a controlled environment to carry out key findings on the development of the child learning process. Experiment results show that the TUIO sessions had 44% faster task completion and 65% fewer errors compared to the gesture-based sessions. The study highlights the effectiveness of TUIOs in child learning environments in the context of interactive system design.**

***Keywords*—Human Computer Interaction, Human Center Design, Tangible User Interface Objects, Hand Gestures, Animal Habitats**

I. Introduction

In recent years, several advanced technologies have been integrated into educational tools, which has significantly changed the way children engage with learning. Interactive systems that leverage gesture recognition, tangible user interface objects (TUIOs), and object detection have emerged as powerful means to create engaging, intuitive and educational experiences. These systems follow the principles of human computer interaction (HCI), by emphasizing on the interface being natural and user-friendly for young children. A user-centered design approach is prioritized by the principles of HCI, which focus on aspects such as responsiveness, learnability, and usability to provide a more engaging user experience [1]. Gesture-based interaction has been quite effective in improving children's learning experiences by making it more interesting and interactive, where the use of recognition systems in various educational games as cognitive-imaginative skills of children is elicited through intuitive and playful interactions [2]. Furthermore, it has also shown that children's gesture education games based on AI gesture recognition technology can strengthen the communication between parents and children, promote emotional exchange. Similarly, hand gesture-based puppetry systems have been utilized to support storytelling with a natural interface that enhances creativity and learning outcomes [3]. Along with gesture-based mechanisms, TUIOs have been found in the educational domains keeping the learning tangible. Using TUIOs in teaching tangible shapes to visually impaired children exemplifies that this kind of interface is able to accommodate diverse learners' needs [4]. The TUI allows the child to work with real-world objects that provide instant feedback for enhanced understanding; such system works in improving understanding by using practical methods of learning. Recognizing children's behaviors and interactions is another critical aspect of designing effective educational systems. Systems that incorporate features that detect and adapt to children's gestures, object manipulations, and even facial expressions can

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create personalized and responsive learning experiences. Such adaptive systems have been shown to improve engagement and learning outcomes by catering to the unique needs and preferences of individual users. For example, systems that use facial expression analysis and recognition to assess children's emotions can adjust the difficulty or content of the learning material, ensuring that the experience remains enjoyable and appropriately challenging [5]. This paper introduces an interactive animal habitat simulator, named *TALH* designed for children in preschool. *TALH* combines gesture recognition, TUIO-based interaction, and object detection to provide an immersive and educational experience. Children can explore animals in their natural habitats, interacting with a circular menu [6] shown in Figure 2 that offers information on each animal's name, sound, description, and habitat. By detecting and responding to children's interactions, the system fosters an intuitive and engaging learning experience.

II. Related Work

A. TUIO-Based Learning

The study [7] investigates how TUIOs and Multitouch Interfaces (MTIs) affect mathematical learning in preschoolers through an especially designed application, called THE NUMBERS. Researchers applied a multi-faceted assessment approach as it included task performance, eye-tracking, and user perceptions. Eye-tracking demonstrated that tasks proposed with TUIOs demanded more cognitive alignment and longer attention to the key instructional material. Feedback revealed that 62.5% of the children preferred TUIOs since it was more fun and more motivating. MTIs, on the other hand, require more precision but on the side of higher levels of stress and less fun. Another study showed that TUIOs enhance peer interaction through physical manipulation, improving outcomes as letter-sound knowledge (PhonoBlocks) and task accuracy (BlackBlock) [8]. Short-term studies report tangible interfaces outperforming traditional systems. Tactile quiz improved accuracy by 12% in 18-minute sessions, while TUIO-based programming tasks reduced errors by 15%. Longitudinal insights, however, remain scarce. Results have shown that TUIOs fared very well for performance metrics. Children using TUIOs completed the tasks 30% quicker and with 25% fewer errors compared to non-tangible systems.

B. Gesture-Based Learning

The study [9] investigates the influence of typical and atypical gestures on narrative comprehension by focusing on its interaction with other variables, including age and task difficulty. Results have shown that participants in the typical gesture condition outperformed participants in the other two conditions significantly on gesture-related comprehension questions. Semantically congruent gestures augment narrative comprehension, improving semantic integration without merely acting through mechanisms that attract attention. It was shown that typical gestures were perceived to be significantly more semantically congruent with the narrative content compared to atypical gestures (mean ratings: 66.98 vs. 33.55, $p < 0.0005$). A children's educational game was proposed in [2], integrating AI gesture recognition into the play of hand-shadows. It designs gesture recognition, animation, an AR photo function, and a question-answering system based on the Unity engine and SenseAR plug-in. Quantitatively, the recognition enhanced cognitive engagements and retention. The poor performance constraint of limiting the designed gesture library was seen, hence needs extension for enhancing various educative scenarios. This study shows that the effectiveness of merging gesture recognition and AR into a serious game in a holistic approach towards interactive and educative play.

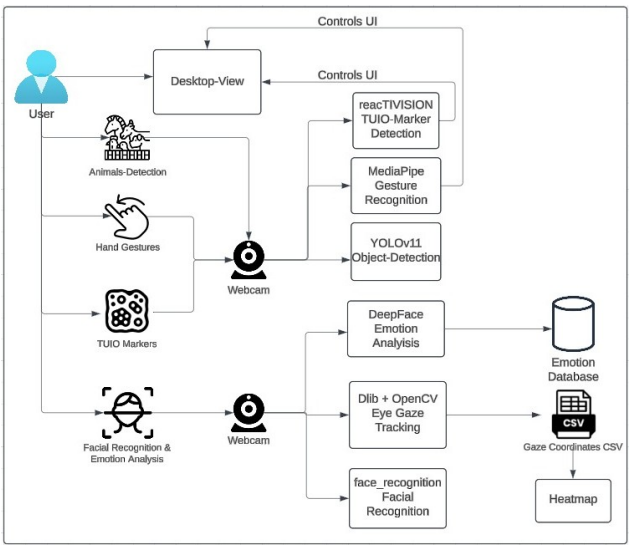

Fig. 1. TALH System Architecture

Fig. 2. Animal in habitat along circular menu

III. System Design and Setup

In this experiment, *TALH*'s system design focuses on delivering an engaging learning experience for preschool children using two interaction methods: TUIOs and hand gestures. Both sessions utilize innovative hardware and software configurations for real-time interaction, tracking, and feedback.

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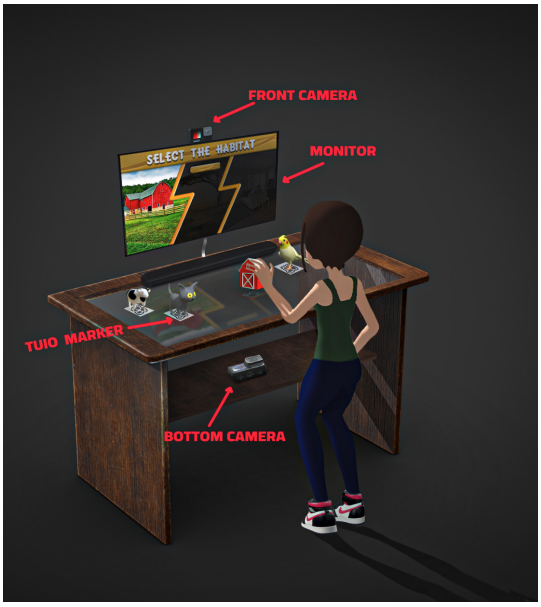

Fig. 3. TUIO System Setup

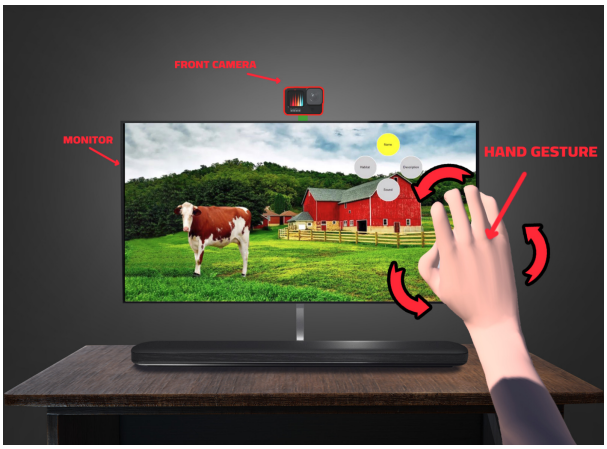

Fig. 4. Hand gestures System Setup

A. TUIO Session

For the TUIO session, the tabletop system was designed using *reacTIVISION* [10], a computer vision framework for object tracking and simple multi-touch, in compliance with Kaltenbrunner [11]. The setup shown in Figure 3 consisted of a table with a 50 cm × 50 cm work area and transparent glass panel on top. To act as an input sensor by wirelessly tracking the movements of tangible animals and habitats. A mobile phone camera was mounted on a mini-tripod beneath the table. The tangible interaction table incorporates reacTIVISION-compatible markers and RFID readers for the detection of tangible animals and habitats. The reacTIVISION framework detects, identifies these markers, and dynamically activates habitats and animals on the screen. While another frontal webcam is responsible for face recognition, emotion analysis, and eye-gaze tracking.

B. Hand Gestures Session

In the hand gestures-based session, one single frontal camera manages all input processing on the tabletop system as shown in Figure 4. Animals is detected using the *YOLOv11* model [12] trained on the COCO dataset [13] containing 10 classes for our chosen animals. The same tangible animals used in the TUIOs session are used for detection by the *YOLOv11* model. Hand gestures is the main critical input for interaction in this session as typical hand gestures were designed for navigation, selection of the desired habitat, and also rotation in the circular menu.

IV. Experiment Design And Analysis

The objective of this research is to conduct a comparative study consisting of two interactive learning sessions, one based on TUIO markers and the other on hand gestures. Each session consisted of learning about animals in three types of habitats, which are wild-life habitat, home habitat, and farm habitat. The system architecture for *TALH* can be shown in Figure 1.

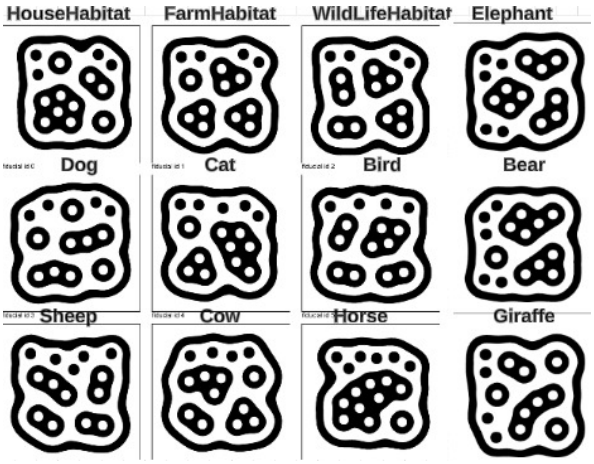

Fig. 5. TUIO Markers and their corresponding labels

A. Tangible User Interface Objects

The TUI functionality developed in *TALH* offers the ability of using reacTIVISION-physical markers as tangible controllers. These markers shall allow children to perform interactive and educational acts through manipulation of real objects on an interactive surface. The setup contains three habitat markers, and nine animal markers representing animals that would belong to such habitats. The TUIO markers mapping can be shown in Figure 5.

B. Hand Gesture Recognition

To enable intuitive user interaction, hand gesture recognition is implemented in *TALH* using a camera placed directly in front of the child to detect hand gestures. The set of gestures used is illustrated in Figure 6. We leverage the *MediaPipe* framework [14] to extract hand landmarks from these gestures, providing a robust set of features for classification. The dataset used to train the model consists of hand landmark data

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Model	Accuracy	AUC	Recall	Precision	F1 Score
Logistic Regression	99.2%	99.4%	99.1%	99.2%	99.5%
K-Nearest-Neighbors (KNN)	98.9%	99.0%	98.7%	98.8%	98.9%
Support Vector Machine (SVM)	98.7%	98.9%	98.5%	98.7%	98.8%
Ridge Classifier	98.6%	98.7%	98.4%	98.5%	98.6%
Random Forest	98.4%	98.8%	98.3%	98.5%	98.7%
Quadratic Discriminant Analysis	98.2%	98.5%	98.0%	98.2%	98.3%
Gradient Boosting	98.1%	98.4%	97.9%	98.1%	98.2%
Linear Discriminant Analysis	97.9%	98.2%	97.8%	98.0%	98.1%
Extra Trees	97.8%	98.0%	97.6%	97.8%	98.0%
Light Gradient Boosting Machine	97.7%	97.9%	97.5%	97.7%	97.8%
Naive Bayes	96.5%	96.8%	96.3%	96.5%	96.6%
Decision Tree	95.8%	96.1%	95.5%	95.8%	96.0%
Extreme Gradient Boosting	95.3%	95.6%	95.0%	95.3%	95.4%
Ada Boost	89.4%	90.2%	88.9%	89.5%	90.0%
Dummy Classifier	50.0%	50.0%	50.0%	50.0%	50.0%

TABLE I
COMPARISON OF CLASSIFICATION MODELS FOR DIFFERENT HAND GESTURES

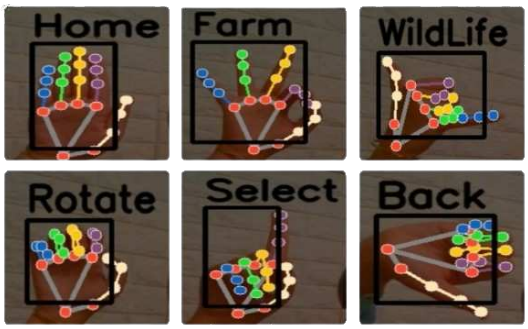

Fig. 6. Hand Gestures and their corresponding labels

extracted from pre-recorded videos capturing each distinct gesture. These features are then fed into a *Logistic Regression* model [15], trained to recognize gestures corresponding to the distinct habitats outlined in the TUIOs section. After tuning various hyperparameters and comparing multiple machine learning models, as shown in Table I, the *Logistic Regression* model achieved a perfect accuracy of 99.2%, demonstrating the effectiveness of both the landmark extraction process and the classification model.

C. Eye Gaze Tracking

Eye gaze tracking was implemented using *OpenCV* and *dlib* to capture the x and y coordinates of the child's eye pupils. These coordinates provided insight into points of interest, such as whether the child focused on the animals or the circular menu. The captured coordinates were aggregated in a CSV file, and a heatmap was generated at the session's end to visualize gaze patterns and assess child engagement.

D. Facial Recognition

Facial recognition feature was built in *TALH* using the *face_recognition* library to personalize interactions. This feature enabled the system to identify children, adapt new learning experiences, and track engagement after identifying the child, helping teachers to gain insight into different changes in emotions in different attempts.

E. Facial Expressions

For facial expression analysis, the *deepface* library [16] was integrated in *TALH* to detect facial expressions concurrently with system operations triggered by markers or gestures. Emotions of interest were happy, sad, fear, surprise, and neutral. These emotions were tracked in real time. The dominant emotion of the session was saved in the end to a *SQLite* database for further teacher reporting analysis.

F. Participants

Children participants aged 3 to 5 years were recruited for this study. Ten children were selected with the prior consent of their parents to perform the experiments. The experiments were aimed at evaluating the effectiveness of TUIOs and hand gesture recognition-based methods.

G. Procedure

The purpose of this experiment was explained to each child, and they were taught how to use the system, including rotation of the markers and understanding the different kinds of gestures available in the game, and how they affect interface control. Afterwards, the child places animals into their correct habitats. This required to use different gestures or TUIOs to navigate through different habitats. The system recognizes such gestures and TUIOs, browsing through different habitats, and rotation & selection of items in the circular menu. The system immediately responds with sound and visual means to make the learning process more interactive and appealing. The experimenters were present throughout the process and were ready to assist the participants whenever needed. Simultaneously, the time taken for each participant to complete each task was tracked using a stopwatch, and then at the end of the session, the teacher opens the system to view the overall report, retrieved from the emotion database.

V. EXPERIMENT RESULTS AND DISCUSSION

In this experiment, tests were conducted to observe and analyze the learning effect among the participants as they interacted with both interaction sessions. Each child played both sessions for three attempts. The time taken to complete the session and the number of errors made were monitored and recorded. Tables II and III show the length in minutes of each child's playtime in three attempts. While tables IV and V present the number of errors made by each child during the three attempts in the two sessions, these errors were defined as instances where the child either misidentified the correct habitat for a given animal or used the incorrect gesture for the desired action, often resulting in confusion or delays in completing the task.

Experimental findings demonstrated that TUIOs and gesture-based systems fostered learning improvements, with quantifiable gains across the attempts. On average, TUIO sessions saw completion times decrease by 25% (from 7.1 to 5.3 minutes) and errors reduced by 55% across the three attempts (Tables II, IV). In contrast, gesture interactions showed slower improvement as completion times decreased by 17% (11.5 to 9.5 minutes) with a 24% error reduction as

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shown in (Tables III, V). TUIOs consistently outperformed gestures, with 32% faster final-attempt times (5.3 vs. 9.5 minutes) and 65% fewer errors. While gesture systems showed promise, notably participant 7 reducing errors from 5 to 4, they exhibited higher initial friction: first-attempt gesture errors were 65% greater than TUIOs. TUIOs' physicality supported steadier progress, with 8/10 children achieving ≤ 2 errors by their third attempt versus 2/10 in gestures. These metrics align with observed cognitive engagement, as TUIOs enabled earlier task mastery despite gesture systems becoming more dynamic with familiarity.

TABLE II
PARTICIPANTS PLAYTIME (IN MINUTES) OVER 3 ATTEMPTS IN TUIO MARKERS SESSION

Kid	First Attempt	Second Attempt	Third Attempt
1	7	8	5
2	9	7	6
3	6	4	5
4	8	7	6
5	6	5	5
6	7	5	5
7	6	5	3
8	7	6	5
9	8	6	4
10	8	9	6

TABLE III
PARTICIPANTS PLAYTIME (IN MINUTES) OVER 3 ATTEMPTS IN HAND GESTURES SESSION

Kid	First Attempt	Second Attempt	Third Attempt
1	12	10	7
2	13	12	9
3	10	10	9
4	14	12	10
5	9	8	8
6	14	12	11
7	8	7	6
8	13	11	10
9	12	9	8
10	12	13	11

TABLE IV
PARTICIPANTS ERROR COUNT OVER 3 ATTEMPTS OVER 3 ATTEMPTS IN TUIOS SESSION

Kid	First Attempt	Second Attempt	Third Attempt
1	5	2	1
2	6	4	3
3	7	5	6
4	5	3	2
5	6	4	3
6	4	3	2
7	5	2	1
8	6	4	1
9	3	5	2
10	5	4	2

VI. CONCLUSION

To conclude, this paper presents a comparative study on the TUIOs and gesture-based interaction techniques in teaching preschool children about animals and their habitats. TALH presents an interactive animal habitat simulator developed to evaluate how such methods influence learning, engagement, and usability in early education. The results have shown the potential of TUIOs to improve cognitive-affective engagement due to the tangible manipulation of objects, which enhances curiosity, motivation, and spatial understanding. Although gesture-based interactions have been proven effective in certain contexts, their usability might need further refinement for young learners. This study in general points to the need for child-centred design of interactive systems that effectively integrate physical and digital elements in the creation of engaging educational experiences.

TABLE V
PARTICIPANTS ERROR COUNT OVER 3 ATTEMPTS IN HAND GESTURES SESSION

Kid	First Attempt	Second Attempt	Third Attempt
1	8	5	3
2	9	8	6
3	10	11	9
4	10	11	9
5	7	7	6
6	10	8	6
7	5	5	4
8	8	6	7
9	12	10	7
10	7	10	8

VII. LIMITATIONS AND FUTURE WORK

Future work should focus on scaling the system for larger groups, exploring diverse educational content, and refining gesture-based interactions to further enhance their effectiveness and usability in preschool learning environments. We suggest extending the eye gaze tracking functionality by advancing it and enabling animal or habitat-based gaze patterns tracking. Such improvements would provide richer, and more immersive experiences for young learners and more feedback to teachers to assess learning processes.

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